QVEX-NGC6503 — Quantum-Hydrodynamic Compressive Barrier: Complete Mathematical and Physical Formulation (English)

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Contents

1 Abstract **1**

2 Introduction: Placement and Scope in the QVEX-NGC6503 Project **1**

3 Methods: Derivation of the Vacuum Bulk Modulus and Compressive Potential **2**

3.1 Vacuum bulk modulus 2

4 Results: Effective Rotation Velocity Profile and Diagnostics **3**

4.1 Effective circular velocity 3

4.2 Unit conversion requirement 3

4.3 Expected diagnostic impact 3

5 Discussion: Boundary Conditions, Anisotropy Bounds, and Consistency with Annex A2 **4**

6 Implementation and Validation Protocol **4**

6.1 Implementation in `nodes/active_tensor_foam.py` 4

6.2 Fitting protocol (`tests/fit_sparc_ngc6503_active.py`) 5

6.3 Test assertions (`tests/test_ngc6503_vacuum_nodes.py`) 5

7 References **6**

1 Abstract

This document delivers the complete English mathematical and physical formulation of the Quantum-Hydrodynamic Compressive Barrier (V_{comp}) for the QVEX-NGC6503 project, together with its exact integration paths, as documented in the Node 05 exchange-current specification, the Annex A2 popcorn boundary conditions, and the ActiveTensorFoamSolver module. The formulation derives a

vacuum bulk modulus $\mathcal{B}$ and an effective potential $V_{\text{eff}}(r) = V_{\text{grav}}(r) + V_{\text{comp}}(r)$ from the canonical vacuum popcorn density law of Annex A2 and the exchange-current framework of Node 05. It specifies the placement of the new theory in `nodes_analysis/05_Vacuum_Exchange_Current_Node.md` (§4) (see S1) and `nodes_analysis/Annex_A2_Vacuum_Popcorn_Boundary_Conditions.md` (§2.5) (see S2), the computational extension of `ActiveTensorFoamSolver` in `nodes/active_tensor_foam.py` (see S3), and the fitting and pytest protocol anchored on `RC_MRT.mrt` data (see S3). All default parameters ($\rho_0 = 7.02 \times 10^{-30}$ g/cm^3, $r_{\text{core}} = 4.5$ kpc, $\lambda_{\text{coh}} = 12.0$ kpc, $\gamma_{\text{vac}} = 1.0$, $R_{\text{cutoff}} = 50.0$ kpc, $\alpha = 0.8$, $G = 4.30091 \times 10^{-6}$) are carried through unchanged, per S3, and the epistemic status of the new terms is declared as an analytical hypothesis until constrained by independent rotation-curve data, following the declaration in S2.

2 Introduction: Placement and Scope in the QVEX-NGC6503 Project

The compressive barrier is integrated at four locations, superseding the placement query resolved in the project review, per S1, S2, and S3: (1) **Theoretical documentation** — a new §4 ("Vacuum Bulk Modulus and Effective Potential") appended to `nodes_analysis/05_Vacuum_Exchange_Current_Node.md` (see S1), and a new §2.5 ("Hydrodynamic Compressive Barrier and Radial Pressure Continuity") appended to `nodes_analysis/Annex_A2_Vacuum_Popcorn_Boundary_Conditions.md` (QVEX-ANNEX-A-BC-01) (see S2); (2) **Computational implementation** — two new methods on the `@dataclass ActiveTensorFoamSolver` in `nodes/active_tensor_foam.py` (see S3): `compressive_potential(r)` and `effective_circular_velocity(r)`, alongside the density unit conversion from g/cm³ to $M_\odot$/kpc^3 that the existing `effective_enclosed_mass` helper currently lacks (a documented, untested limitation in S3); (3) **Fitting** — extension of `tests/fit_sparc_ngc6503_active.py`, which drives the solver against SPARC `RC_MRT.mrt` data (see S3); (4) **Testing** — new assertions in `tests/test_ngc6503_vacuum_nodes.py`, which already parses Node equations from Markdown and cross-checks them against the numeric module, per S1 and S3.

The formulation is deliberately conservative: it reuses the already-established popcorn density law of Annex A2 §3 (S2) and the exchange current $\Gamma(r) \geq 0$ of Node 05 (S1), and introduces no new free parameter beyond the vacuum adiabatic coefficient γ_{vac}, which is fixed at 1.0 in the codebase (S3). The baseline goodness-of-fit diagnostics against which any improvement is judged are $\chi^2_{\text{total}} = 58346.27$, $\chi^2_{\text{reduced}} = 1882.14$ with dof = 31, evaluated for the NGC 6503 rotation curve; the PID controller gains $K_P = 0.92$, $K_I = 0.35$, $K_D = 0.18$ remain untouched (S3).

3 Methods: Derivation of the Vacuum Bulk Modulus and Compressive Potential

3.1 Vacuum bulk modulus

For a barotropic vacuum fluid with adiabatic coefficient $\gamma_{\rm vac}$, the equation of state is $p(\rho) = (\gamma_{\rm vac} - 1)\,\rho c^2$ and the bulk modulus is the thermodynamic stiffness:

$$\mathcal{B} \equiv \rho \frac{\partial p}{\partial \rho} = \gamma_{\rm vac}\,\rho(r)\,c^2.$$

With $\gamma_{\rm vac} = 1.0$ this reduces to the stiff-fluid (Zel'dovich) form $\mathcal{B} = \rho c^2$, i.e., the vacuum resists compression with sound speed $c_s = \sqrt{\mathcal{B}/\rho} = c$ — the maximal causal stiffness consistent with the microcausality constraint $\xi(r) \le R_{\rm horizon}(r)$ of Annex A2 §2.2 [S2]. This is the physically motivated choice: the barrier can never transmit pressure information superluminally, and $\mathcal{B}(r)$ inherits the spatial falloff of $\rho_{\rm popcorn}(r)$, vanishing as $r \to \infty$ in accord with the scale-separation and screening constraint $\lim_{r\to\infty} \rho = 0$ of Annex A2 §2.3 [S2]. The parameter bounds that the barrier must respect are exactly those of Annex A2 §3: $0 \le \rho_0 \le 10^{-25}\,\mathrm{kg/m^3}$ strictly non-negative, $r_s > 0$, $r_c > r_s$ (or $r_c \to \infty$), and $0 \le \alpha \le 4$ [S2]; in the codebase the realized values $\rho_0 = 7.02\times10^{-30}\,\mathrm{g/cm^3}$, $r_s = 4.5$ kpc, $r_c = 50.0$ kpc, $\gamma_{\rm vac} = 1.0$ all lie inside these bounds [S3].

4 Results: Effective Rotation Velocity Profile and Diagnostics

4.1 Effective circular velocity

With the effective potential $V_{\rm eff}$, the circular velocity of a test mass is obtained from the gradient of the total potential. With $G = 4.30091 \times 10^{-6}\ \mathrm{kpc\,(km/s)^2}\,M_\odot^{-1}$ [S3],

$$v_c^2(r) \;=\; r\,\frac{\partial}{\partial r}\left[V_{\rm grav}(r) + V_{\rm comp}(r)\right] \;=\; \frac{G\,M_{\rm enc}(r)}{r}\,S(r) \;+\; \frac{\gamma_{\rm vac}}{c^2}\,r\,\frac{\partial}{\partial r}\left[\rho_{\rm popcorn}(r)\,\Phi_N(r)^2\right],$$

where $S(r) = 1/(1 + (r/\lambda_{\rm coh})^\alpha)$ is the existing screening factor ($S(0) = 1$, $S(\lambda_{\rm coh}) = 0.5$, $S(\infty) \to 0$) [S3]. Expanding the barrier term for the code profile with $\gamma_{\rm vac} = 1.0$ [S3]:

$$\frac{\partial}{\partial r}\left[\rho\,\Phi_N^2\right] = \rho\left(2\Phi_N\,\Phi_N' + \Phi_N^2\,\frac{\rho'}{\rho}\right), \qquad \frac{\rho'}{\rho} = -\frac{\gamma_{\rm vac}}{r_s + r} - \frac{1}{r_c} = -\frac{1}{r_{\rm core} + r} - \frac{1}{R_{\rm cutoff}}.$$

The second term of ρ'/ρ is strictly negative, so the barrier contribution to v_c^2 can be positive (where $\Phi_N' > 0$ and the density gradient term is subdominant), producing a *coherent additional lift* to the outer rotation velocity without any sign flip — consistent with the Node 05 sign-invariance argument at $r \approx 12.11$ kpc [S1].

4.2 Unit conversion requirement

The existing `effective_enclosed_mass` helper feeds ρ_0 in g/cm³ into a solver using G in kpc–$M_\odot$ units — a documented, untested limitation [S3]. The conversion must be made explicit in the new methods:

$$1\ \mathrm{g/cm^3} = \frac{(0.01\ \mathrm{m})^{-3}}{1.98892 \times 10^{30}\ \mathrm{kg}} \ \Rightarrow\ \rho_0 = 7.02 \times 10^{-30}\ \mathrm{g/cm^3} \approx 1.05 \times 10^{2}\ M_\odot\,\mathrm{kpc}^{-3}.$$

Using ρ_{popcorn} in $M_\odot/\mathrm{kpc}^3$ inside V_{comp} keeps the barrier term in units of $(\mathrm{km/s})^2$ when multiplied by $r\,\partial_r$ with Φ_N in $(\mathrm{km/s})^2$.

4.3 Expected diagnostic impact

The barrier term is exponentially suppressed for $r \gg R_{\mathrm{cutoff}}$ and modest in the inner disk, so the predicted change is a targeted improvement of the outer-residual chi-square contribution while leaving the fitted inner profile and the published metrics $\chi^2_{\mathrm{total}} = 58346.27$, $\chi^2_{\mathrm{reduced}} = 1882.14$ (dof = 31) as the comparison baseline [S3]. The SPARC driver `tests/fit_sparc_ngc6503_active.py` should be re-run in two modes (barrier off / barrier on) and the $\Delta\chi^2$ per radial bin reported, since Node 05 explicitly warns that removing the legacy domain penalty requires re-running any fit that relied on it [S1].

5 Discussion: Boundary Conditions, Anisotropy Bounds, and Consistency with Annex A2

The compressive barrier must satisfy the hard analytical constraints of Annex A2 (§2, document QVEX-ANNEX-A-BC-01) [S2], and it does so structurally:

1. **Bianchi identity / local conservation** ($\nabla^\mu T^{(\mathrm{popcorn})}_{\mu\nu} = 0$, Annex A2 §2.1) [S2]. V_{comp} is derived from a barotropic equation of state $p = (\gamma_{\mathrm{vac}} - 1)\rho c^2$; the resulting pressure-gradient force is conservative and the exchange pair (matter source / vacuum sink) sums exactly to zero, as already machine-checked for Node 05 [S1]. No net four-momentum is created.
2. **Microcausality** (Annex A2 §2.2) [S2]. With $\gamma_{\mathrm{vac}} = 1.0$, $c_s = c$ exactly, so pressure perturbations never propagate superluminally, and the correlation length of the coherent bubbles obeys $\xi(r) \le R_{\mathrm{horizon}}(r)$.
3. **Radial pressure continuity** (the content of new §2.5 in Annex A2) [S2]. At the popcorn boundary the radial pressure must be continuous: $[p_r]^+_- = 0$, i.e. $p_r^{\mathrm{out}} - p_r^{\mathrm{in}} = 0$ across any shell where the effective potential is matched. Since $V_{\mathrm{comp}} \propto \rho\,\Phi_N^2$ and both ρ and Φ_N are continuous for a finite baryonic disk, the barrier introduces no discontinuity in p_r.

4. **Anisotropy bound** (Annex A2 §2.4) [S2]. The barrier is isotropic in the vacuum rest frame (a scalar bulk response); it adds nothing to $|p_r - p_t|$, so the bound $|p_r - p_t|/(\rho c^2) \leq \Gamma_{\max}$ is preserved by construction.
5. **Screening / asymptotic decay** (Annex A2 §2.3) [S2]. $V_{\rm comp}(r) \to 0$ as $r \to \infty$, at least as fast as $\rho_{\rm popcorn} \sim e^{-r/R_{\rm cutoff}}$ [S3], so the outer halo remains unscreened by the barrier at large radii.

Limitations, declared explicitly in the Node 05 style. (i) $\varepsilon(r) \simeq \Phi_N/c^2$ is a leading-order Newtonian response; no field equation is derived, and $\Gamma(r)$ remains calibrated rather than derived, exactly as stated in the "What This Node Does Not Claim" section of Node 05 [S1]. (ii) The epistemic status of the barrier is *Unverified / Analytical Hypothesis Boundary Condition*, matching the header declaration of Annex A2 (QVEX-ANNEX-A-BC-01) [S2]; it becomes physical only after independent constraint by rotation-curve, lensing, or CMB data. (iii) The symbolic duality between Annex A2's α (density exponent) and the code's `alpha = 0.8` (screening exponent) must be documented in §2.5 to prevent misconfiguration [S2, S3].

6 Implementation and Validation Protocol

6.1 Implementation in `nodes/active_tensor_foam.py`

Two methods are added to the `@dataclass ActiveTensorFoamSolver` [S3], keeping all existing defaults untouched ($\rho_0 = 7.02 \times 10^{-30}$, `lambda_coh = 12.0`, `gamma_vac = 1.0`, `R_cutoff = 50.0`, `friction_scale = 1.1`, `r_core = 4.5`, `alpha = 0.8`, `G = 4.30091e-6`) [S3]:

```
# Unit conversion (module-level constant)
G_PER_CM3_TO_MSUN_PER_KPC3 = 1.0 / (1.98892e33 * (3.08567758e21)**-3 * 1.0)  # calibrate in t

def compressive_potential(self, r, phi_n):
    """V_comp = (gamma_vac / (2 c^2)) * rho(r) * Phi_N^2, (km/s)^2 units."""
    r = self._nonnegative_radius(r)
    rho = self.vacuum_density(r) * G_PER_CM3_TO_MSUN_PER_KPC3
    return 0.5 * self.gamma_vac * rho * phi_n**2 / C_KMS**2

def effective_circular_velocity(self, r, m_baryon):
    """v_eff^2 = G M_enc S(r)/r + r d/dr[V_comp]; r=0 guarded, clipped >= 0."""
    ...  # finite-difference d/dr of compressive_potential, add to circular_velocity
```

Both methods must (a) route `r` through the module helper `_nonnegative_radius` so a negative or non-finite radius raises `ValueError` rather than being clipped (Annex A, A.3.1.2 convention) [S1, S3], and (b) guard `r = 0` with the same 10^{-9}-style epsilon used

in `circular_velocity` and clip `v_eff^2` at zero from below [S3]. The barrier term is appended to the `compute(r)` output dictionary under the key `"compressive_potential"` so the Markdown-parsing test harness can see it [S3].

6.2 Fitting protocol (`tests/fit_sparc_ngc6503_active.py`)

The driver loads `RC_MRT.mrt`, evaluates the baryonic Φ_N per radial bin, and fits in two modes: barrier off (reproducing $\chi^2_{\text{total}} = 58346.27$, $\chi^2_{\text{reduced}} = 1882.14$, dof $= 31$) and barrier on [S3]. Reported diagnostics: per-bin $\Delta\chi^2$, total $\Delta\chi^2$ with dof unchanged (no new free parameter, since $\gamma_{\text{vac}} = 1.0$ is fixed), and the PID stability check with gains $K_P = 0.92$, $K_I = 0.35$, $K_D = 0.18$ [S3].

6.3 Test assertions (`tests/test_ngc6503_vacuum_nodes.py`)

Following the existing pattern of parsing equations from the Markdown nodes and comparing against the numeric module [S3], the new assertions are:

- $V_{\text{comp}}(r) \geq 0$ for all $r \geq 0$ (barrier property).
- $V_{\text{comp}}(r) \to 0$ within machine tolerance for $r > 10\,R_{\text{cutoff}}$ (screening).
- Continuity: $|V_{\text{comp}}(r^-) - V_{\text{comp}}(r^+)| \to 0$ across sampled shell boundaries (radial pressure continuity, Annex A2 §2.5) [S2].
- Unit conversion: `vacuum_density(0) * G_PER_CM3_TO_MSUN_PER_KPC3` $\approx$ `1.05e2 M_sun/kpc^3` within 0.5% (closes the documented gap in `effective_enclosed_mass`) [S3].
- Radius guard: negative and infinite radii raise `ValueError` for both new methods [S1].
- χ^2_{reduced} (barrier on) $\leq$ χ^2_{reduced} (barrier off) for the $R > R_{\text{cutoff}}$ bin subset of `RC_MRT.mrt`.

Run with `python -m pytest -v` from the repository root, per the PowerShell instruction already recorded in Annex A2 §4 [S2].

7 References

- [S1] QVEX-NGC6503-Quantum-Vacuum-(9).zip — `nodes_analysis/05_Vacuum_Exchange_Curren` (Node 05: Causal Exchange Current), lines 1–80.
- [S2] QVEX-NGC6503-Quantum-Vacuum-(9).zip — `nodes_analysis/Annex_A2_Vacuum_Popcorn_B` (QVEX-ANNEX-A-BC-01), lines 1–98.
- [S3] QVEX-NGC6503-Quantum-Vacuum-(9).zip — `nodes/active_tensor_foam.py` (`ActiveTensorFoamSolver`), lines 1–186.